

# Lab 4.1: Securing VPC Resources by Using Security Groups

Security groups are virtual firewalls that are attached to Amazon Elastic Compute Cloud (Amazon EC2) instances. Security group rules define what traffic is allowed in or out of an instance. In this lab, you are tasked to configure access rules for EC2 instances.

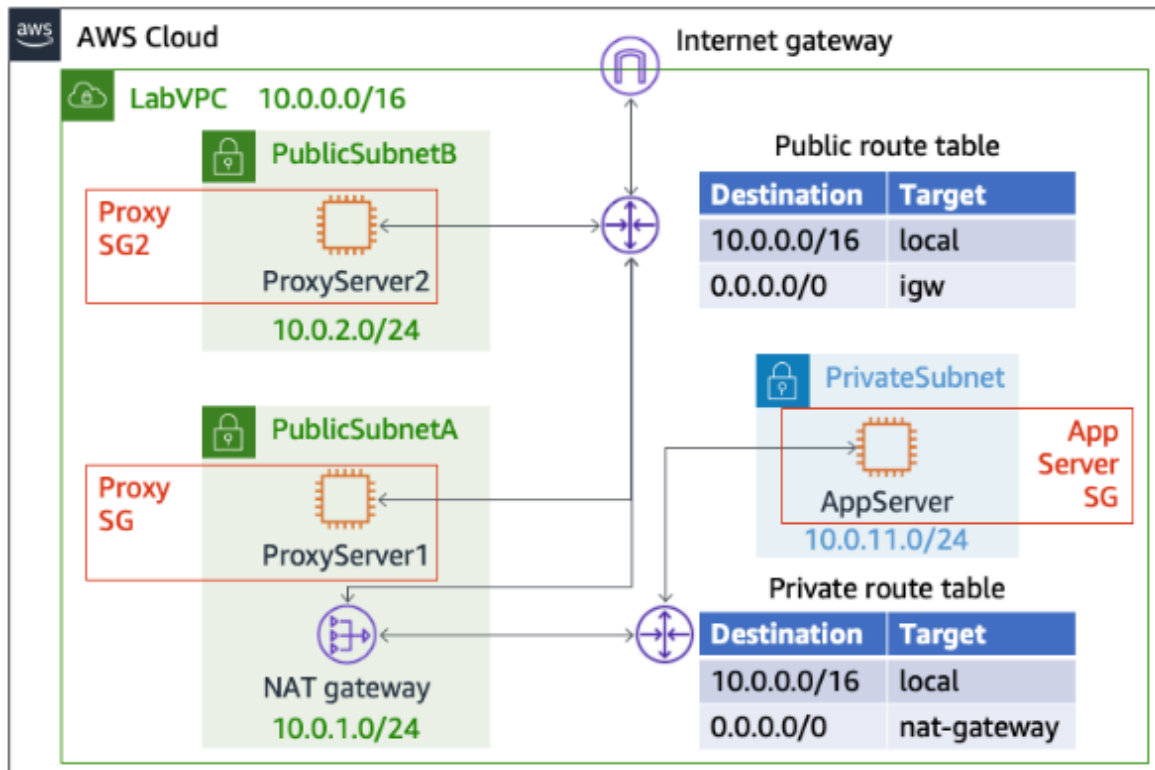
After completing this lab, you should be able to do the following:

- Examine security groups to determine what traffic is allowed.
- Change which security groups are applied to EC2 instances.
- Create new security groups.
- Update the inbound rules on security groups to follow the principle of least privilege.
- Understand how security groups can reference other security groups.
- Configure a network access control list (ACL) to block traffic on a specific TCP port.
- Connect to an instance in a private subnet by using SSH.
- Connect to an instance in a private subnet by using AWS Systems Manager Session Manager.

## Scenario

The lab environment includes a fully functional virtual private cloud (VPC) with multiple subnets, route tables, an internet gateway, a NAT gateway, and multiple security groups. In addition, three EC2 Linux instances have been launched into different subnets: two into public subnets and one into a private subnet. The instances in the public subnets are configured as proxy servers to forward traffic to the application server in the private subnet. You will connect to the proxy instances by using a web browser to test connectivity to the private application server to verify how security groups restrict access.

The following diagram shows the architecture that was created for you in the AWS account at the beginning of the lab.



## Task 1: Analyzing the VPC and private subnet resource settings

In this first task, you will explore the resources configured in the LabVPC, so that you will be familiar with its current network settings.

In the AWS Management Console, in the search box to the right of **Services**, search for and choose **VPC** to open the Amazon Virtual Private Cloud (Amazon VPC) console. Analyze the LabVPC and the subnets configured in it.

- In the navigation pane, choose **Your VPCs**.

The AWS Region that you are currently observing has two VPCs. One is the default VPC, which has a CIDR range of 172.31.0.0/16. The other is named **LabVPC** and has a CIDR range of 10.0.0.0/16. You will use the LabVPC in this lab

Sus VPC (2) [Información](#)

Q Buscar VPC por atributo o etiqueta

| <input type="checkbox"/> | Name   | ID de la VPC                          | Estado    | Bloquear el ... | CIDR IPv4     | CIDR IPv6 |
|--------------------------|--------|---------------------------------------|-----------|-----------------|---------------|-----------|
| <input type="checkbox"/> | LabVPC | <a href="#">vpc-07b95d79f297b5cf3</a> | Available | Desactivado     | 10.0.0.0/16   | -         |
| <input type="checkbox"/> | -      | <a href="#">vpc-0433ec18e558e99db</a> | Available | Desactivado     | 172.31.0.0/16 | -         |

- In the navigation pane, choose **Subnets**.
- In the subnets list, choose the **VPC** column header to organize the subnets by VPC, as shown in the following image

| Subredes (9) Información                  |               |                                          |           |                                                 |                 |                |
|-------------------------------------------|---------------|------------------------------------------|-----------|-------------------------------------------------|-----------------|----------------|
| Q Buscar subredes por atributo o etiqueta |               |                                          |           |                                                 |                 |                |
| <input type="checkbox"/>                  | Name          | ID de subred                             | Estado    | VPC                                             | Bloquear el ... | CIDR IPv4      |
| <input type="checkbox"/>                  | PublicSubnetA | <a href="#">subnet-0fca9126d857e3418</a> | Available | <a href="#">vpc-07b95d79f297b5cf3   LabV...</a> | Desactivado     | 10.0.1.0/24    |
| <input type="checkbox"/>                  | PrivateSubnet | <a href="#">subnet-0b3a60125e3a1d0b4</a> | Available | <a href="#">vpc-07b95d79f297b5cf3   LabV...</a> | Desactivado     | 10.0.11.0/24   |
| <input type="checkbox"/>                  | PublicSubnetB | <a href="#">subnet-0df1f5d7b43cc19c6</a> | Available | <a href="#">vpc-07b95d79f297b5cf3   LabV...</a> | Desactivado     | 10.0.2.0/24    |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-06577bacdc7b2b8da</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     | 172.31.32.0/20 |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-06049af9b3b681f3b</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     | 172.31.64.0/20 |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-09d675006db9a4126</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     | 172.31.80.0/20 |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-0e34676086fd9f412</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     | 172.31.16.0/20 |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-07bd93e1b6898409a</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     | 172.31.0.0/20  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-09157f273fdfe2291</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     | 172.31.48.0/20 |

Notice that LabVPC has three subnets defined in it. They are named **PrivateSubnet**, **PublicSubnetA**, and **PublicSubnetB**, and each one has a unique CIDR range of private IPv4 addresses assigned to it

Analyze the routing details that apply to EC2 instances in the PrivateSubnet, and change the name of the route table.

- Select the checkbox for **PrivateSubnet**, and then in the lower pane, choose the **Route table** tab.

| Subredes (1/9) Información                |               |                                          |           |                                                 |                 |  |
|-------------------------------------------|---------------|------------------------------------------|-----------|-------------------------------------------------|-----------------|--|
| Q Buscar subredes por atributo o etiqueta |               |                                          |           |                                                 |                 |  |
| <input type="checkbox"/>                  | Name          | ID de subred                             | Estado    | VPC                                             | Bloquear el ... |  |
| <input type="checkbox"/>                  | PublicSubnetA | <a href="#">subnet-0fca9126d857e3418</a> | Available | <a href="#">vpc-07b95d79f297b5cf3   LabV...</a> | Desactivado     |  |
| <input checked="" type="checkbox"/>       | PrivateSubnet | <a href="#">subnet-0b3a60125e3a1d0b4</a> | Available | <a href="#">vpc-07b95d79f297b5cf3   LabV...</a> | Desactivado     |  |
| <input type="checkbox"/>                  | PublicSubnetB | <a href="#">subnet-0df1f5d7b43cc19c6</a> | Available | <a href="#">vpc-07b95d79f297b5cf3   LabV...</a> | Desactivado     |  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-06577bacdc7b2b8da</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     |  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-06049af9b3b681f3b</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     |  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-09d675006db9a4126</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     |  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-0e34676086fd9f412</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     |  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-07bd93e1b6898409a</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     |  |
| <input type="checkbox"/>                  | -             | <a href="#">subnet-09157f273fdfe2291</a> | Available | <a href="#">vpc-0433ec18e558e99db</a>           | Desactivado     |  |

#### subnet-0b3a60125e3a1d0b4 / PrivateSubnet

Detalles | Registros de flujo | **Tabla de enrutamiento** | ACL de red | Reservas de CIDR | Uso compartido | Etiquetas

Tabla de enrutamiento: [rtb-07d41aa63a8f57da0 / changeme](#)

| Rutas (2)       |                                       |
|-----------------|---------------------------------------|
| Q Filtrar rutas |                                       |
| Destino         | Destino                               |
| 10.0.0.0/16     | local                                 |
| 0.0.0.0/0       | <a href="#">nat-095123214b704f58d</a> |

- Above the list of routes, choose the link for the route table that is associated with this subnet.
- Choose the **Routes** tab.

rtb-07d41aa63a8f57da0 / changeme

**Detalles** Información

ID de tabla de enrutamiento  
rtb-07d41aa63a8f57da0

VPC  
vpc-07b95d79f297b5cf3 | LabVPC

Principal  
No

ID de propietario  
137328169203

Asociaciones de subredes explícitas  
subnet-0b3a60125e3a1d0b4 / PrivateSubnet

Asociaciones de borde  
-

**Rutas** Asociaciones de subredes Asociaciones de borde Propagación de rutas Etiquetas

Rutas (2)

Filtrar rutas

| Destino     | Objetivo                              | Estado | Propagada |
|-------------|---------------------------------------|--------|-----------|
| 0.0.0.0/0   | <a href="#">nat-095123214b704f58d</a> | Activo | No        |
| 10.0.0.0/16 | local                                 | Activo | No        |

Notice that all IP addresses in the LabVPC CIDR range (10.0.0.0/16) will be routed to local targets within the VPC. All other traffic (0.0.0.0/0) will be routed to a NAT gateway.

**Note:** If 0.0.0.0/0 traffic for this subnet was instead routed to an internet gateway, then you would know that it is a *public* subnet with direct access to the internet. *Instead*, it routes network traffic to a NAT gateway.

- Choose the **Tags** tab, and then choose **Manage tags**.
- Change the name of the route table from **changeme** to **Private** and then choose **Save**.

Etiquetas actualizadas correctamente

rtb-07d41aa63a8f57da0 / Private

**Detalles** Información

ID de tabla de enrutamiento  
rtb-07d41aa63a8f57da0

VPC  
vpc-07b95d79f297b5cf3 | LabVPC

Principal  
No

ID de propietario  
137328169203

Asociaciones de subredes explícitas  
subnet-0b3a60125e3a1d0b4 / PrivateSubnet

**Rutas** Asociaciones de subredes Asociaciones de borde Propagación de rutas **Etiquetas**

**Etiquetas**

Buscar etiquetas

| Clave           | Valor                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------|
| aws:cloudfor... | PrivateRT                                                                                                                        |
| aws:cloudfor... | c162226a4160672110375184t1w137328169203                                                                                          |
| cloudlab        | c162226a4160672110375184t1w137328169203                                                                                          |
| Name            | Private                                                                                                                          |
| aws:cloudfor... | arn:aws:cloudformation:us-east-1:137328169203:stack/c162226a4160672110375184t1w137328169203/49446fb0-3d82-11f0-bd6b-0e6572e747c3 |

Choose the **Routes** tab, and then choose the **nat-xxxxxx** link, which is the target for non-local traffic.

**Rutas** Asociaciones de subredes Asociaciones de borde Propagación

Rutas (2)

Filtrar rutas

| Destino     | Objetivo                              |
|-------------|---------------------------------------|
| 0.0.0.0/0   | <a href="#">nat-095123214b704f58d</a> |
| 10.0.0.0/16 | local                                 |

The diagram illustrates a network architecture within an AWS Cloud. A VPC (Virtual Private Cloud) is shown with a CIDR block of 10.0.0.0/16. It contains two public subnets: PublicSubnetA (10.0.1.0/24) and PublicSubnetB (10.0.2.0/24). A NAT gateway is located in PublicSubnetA, with a CIDR block of 10.0.1.0/24. An Internet gateway is connected to the VPC. A PrivateSubnet (10.0.11.0/24) contains an AppServer. A Private route table is associated with the PrivateSubnet, showing routes for 10.0.0.0/16 (local) and 0.0.0.0/0 (nat-gateway). The AppServer is connected to the Internet gateway via the NAT gateway.

**Internet gateway**

**LabVPC 10.0.0.0/16**

**PublicSubnetB 10.0.2.0/24**

**PublicSubnetA 10.0.1.0/24**

**NAT gateway 10.0.1.0/24**

**PrivateSubnet 10.0.11.0/24**

**AppServer 10.0.11.0/24**

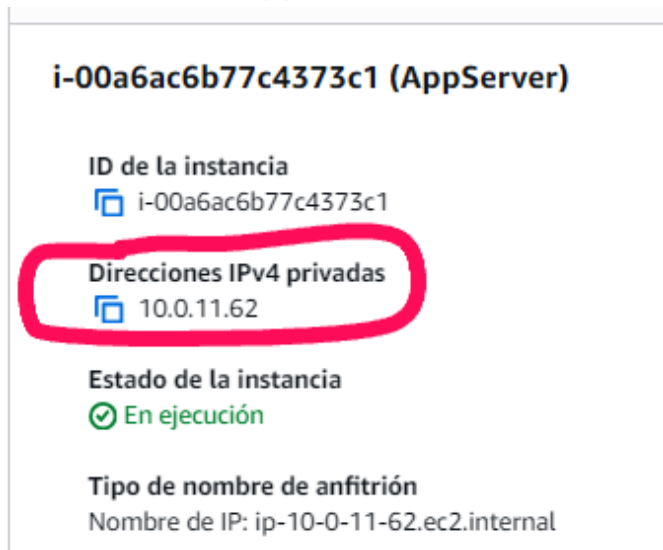
**App Server SG**

**Private route table**

| Destination | Target      |
|-------------|-------------|
| 10.0.0.0/16 | local       |
| 0.0.0.0/0   | nat-gateway |

Observe details about the AppServer EC2 instance, which is in the PrivateSubnet, and its associated security group

- In the console, in the search box to the right of to **Services**, search for and choose **EC2**.
- **Important:** *Right-click* the **EC2** service in the results to open the Amazon EC2 console in a *new browser tab*.
  - **Tip:** Keep the Amazon VPC console and Amazon EC2 console browser tabs open throughout this lab. You will use both of them in later steps.
- In the Amazon EC2 console, in the navigation pane, choose **Instances**
  - Select the **AppServer** instance, and on the **Details** tab:



(We know the instance is running in the private subnet by looking at the IP Address. Remember that the Range for this subnet is 10.0.11.0/24 and the AppServer instance has the IP 10.0.11.62)

- Notice that this instance is running in the PrivateSubnet.
- Also notice that it has a *private* IPv4 address, but it does not have a *public* IPv4 address.
  - **Note:** A *private* IPv4 address is an IP address that's not reachable over the Internet. Private IPv4 addresses can be used for communication between instances in the same VPC.
- Choose the **Security** tab, and then choose the link of the security group that is associated with this instance. The security group name contains **AppServerSG**.

|                                     | Name         | ID de la instancia  | Estado de la i... | Tipo de inst... |
|-------------------------------------|--------------|---------------------|-------------------|-----------------|
| <input checked="" type="checkbox"/> | AppServer    | i-00a6ac6b77c4373c1 | En ejecución      | t2.small        |
| <input type="checkbox"/>            | ProxyServer2 | i-0d7e7b543afed7749 | En ejecución      | t2.small        |
| <input type="checkbox"/>            | ProxyServer1 | i-005ecef81045b42f7 | En ejecución      | t2.small        |

### i-00a6ac6b77c4373c1 (AppServer)

Rol de IAM

[SsmRole](#)

Hora de lanzamiento

Fri May 30 2025 13:19:57 GMT-0500 (hora estándar oriental)

ID del propietario

137328169203

Grupos de seguridad



[sg-0666eaa18124b4a29](#)

[\(c162226a4160672110375184t1w13732816920](#)

[3-AppServerSG-hrfNI91CUAbU\)](#)

- Choose the **Inbound rules** tab, and notice that a rule has been defined. HTTP traffic over TCP port 80 with a source of 0.0.0.0/0 (anywhere) is allowed.
  - Tip:** You might need to scroll the inbound rules table to the right to see all of these details.

## Detalles

### Nombre del grupo de seguridad

 c162226a4160672110375184t1w1373281  
69203-AppServerSG-hrfNI91CUAbU

### ID del grupo de seguridad

 sg-0666eaa18124b4a29

### Descripción

 Security group for App server

### ID de la VPC

[vpc-07b95d79f297b5cf3](#) 

### Propietario

 137328169203

### Número de reglas de entrada

1 Entrada de permiso

### Número de reglas de salida

1 Entrada de permiso

< [Reglas de entrada](#) | Reglas de salida | Compartiendo : *novedad* | Asociar >

## Reglas de entrada (1)



Administrar etiquetas

[Editar reglas de entrada](#)

 Buscar

| < 1 >  |           |                      |           |
|--------------------------------------------------------------------------------------------|-----------|----------------------|-----------|
| Tipo                                                                                       | Protocolo | Intervalo de puertos | Origen    |
| HTTP                                                                                       | TCP       | 80                   | 0.0.0.0/0 |

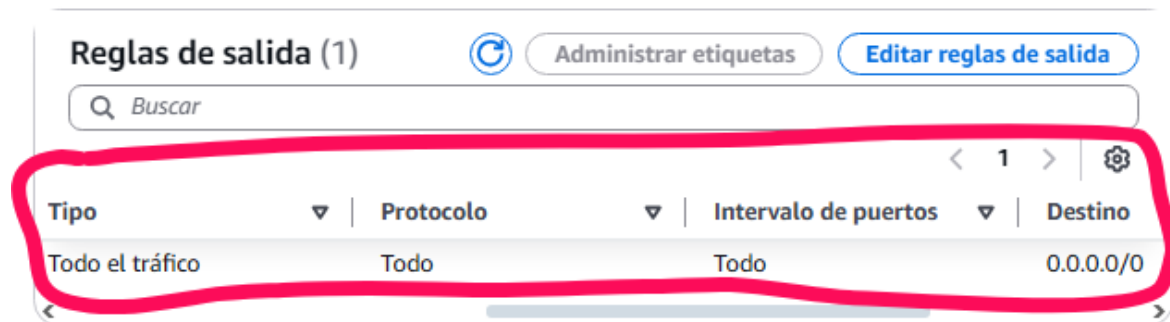
- **Note:**

- This means that any instance that has the AppServerSG security group attached to it can receive connections *from* any computer that attempts to communicate with it on TCP port 80.
- By default, when you create a security group, it does not have any inbound rules, which means that inbound network traffic is not permitted. However, this security group has one inbound rule defined.
- Security groups are *stateful*. This means that if an instance sends a request, the response traffic for that request is allowed in regardless of inbound security group rules.

- Choose the **Outbound rules** tab.

Notice that all outbound traffic is allowed. This is the default setting for security groups.





Now that you have analyzed the private subnet settings, you will look at the same details for the public subnets.

## Task 2: Analyzing the public subnet resource settings

In this task, you will analyze the networking, routing, and security settings in the public subnets of the VPC.

In the Amazon VPC console, analyze the routing details that affect EC2 instances in PublicSubnetA.

- Return to the browser tab where the Amazon VPC console is open.
- In the navigation pane, choose **Subnets**, and then select **PublicSubnetA**.

**Note:** If necessary, clear the checkbox for PrivateSubnet. If more than one item is selected in the list, then the **Details** tab and other tabs will not display below. Keep this in mind throughout the lab.

- Choose the **Route table** tab.

Notice that this public subnet route table routes 0.0.0.0/0 traffic to an internet gateway.



**Note:** This subnet is called a *public* subnet because it is configured to route non-local traffic to an internet gateway.

- Choose the **igw-xxxxxx** link, which is the target for non-local traffic.

On the **Details** tab, notice that this internet gateway is associated with the LabVPC.

**Gateways de Internet (1/1)** Información  Acciones  Crear gateway de Internet

🔍 *Buscar puertas de enlace de Internet por atributo o etiqueta*

ID de gateway de Internet : igw-0ac436c34f5d25840  Borrar filtros < 1 > 

| <input checked="" type="checkbox"/> | Name | ID de gateway de Internet             | Estado                                                                                       |
|-------------------------------------|------|---------------------------------------|----------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | IGW  | <a href="#">igw-0ac436c34f5d25840</a> |  Attached |

## igw-0ac436c34f5d25840 / IGW

Detalles Etiquetas

### Detalles

ID de gateway de Internet

 igw-0ac436c34f5d25840

Estado

 Attached

ID de la VPC

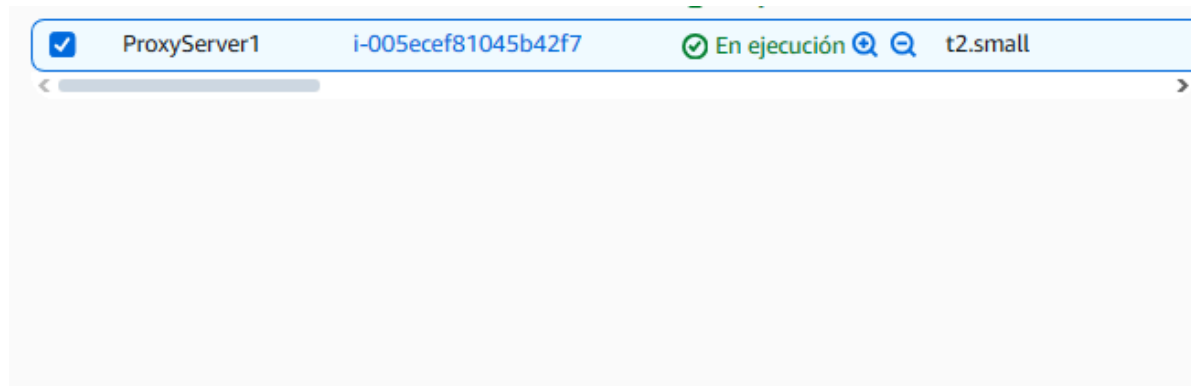
[vpc-07b95d79f297b5cf3](#)  
[LabVPC](#)

Propietario

 137328169203

- Observe details about the ProxyServer1 instance, which is running in PublicSubnetA, and its associated security group.
- Return to the browser tab where the Amazon EC2 console is open.
- In the navigation pane, choose **Instances**.
- Select the **ProxyServer1** instance, and then choose the **Details** tab.

Notice that, unlike the AppServer instance that runs in a *private* subnet, this server runs in a *public* subnet and has a *public* IPv4 address.



### i-005ecef81045b42f7 (ProxyServer1)

#### ID de la instancia

i-005ecef81045b42f7

#### Direcciones IPv4 privadas

10.0.1.33

#### Estado de la instancia

En ejecución

#### Tipo de nombre de anfitrión

Nombre de IP: ip-10-0-1-33.ec2.internal

#### Dirección IPv4 pública

3.90.204.40 | [dirección abierta](#)

#### Dirección IPv6

–

#### DNS público

[ec2-3-90-204-40.compute-1.amazonaws.com](#) | [dirección abierta](#)

#### Nombre DNS de IP privada (solo IPv4)

[ip-10-0-1-33.ec2.internal](#)

Remember that the public subnet has the range 10.0.1.0/24

- Choose the **Security** tab.

Notice that a security group with **ProxySG** in its name is associated with this EC2 instance. As you can see in the **Inbound rules** section, the security group allows inbound traffic on TCP port 80 from any source.



|                                     |              |                     |              |          |
|-------------------------------------|--------------|---------------------|--------------|----------|
| <input checked="" type="checkbox"/> | ProxyServer2 | i-0d7e7b543afed7749 | En ejecución | t2.small |
| <input type="checkbox"/>            | ProxyServer1 | i-005ecef81045b42f7 | En ejecución | t2.small |

## i-0d7e7b543afed7749 (ProxyServer2)

< **Detalles** Estado y alarmas Monitoreo Seguridad Redes >

### ▼ Resumen de instancia Información

#### ID de la instancia

i-0d7e7b543afed7749

#### Direcciones IPv4 privadas

10.0.2.120

#### Estado de la instancia

En ejecución

#### Dirección IPv4 pública

54.196.9.100 | [dirección abierta](#)

#### Dirección IPv6

-

#### DNS público

ec2-54-196-9-100.compute-1.amazonaws.com | [dirección abierta](#)

- Choose the **Security** tab. Notice that a security group with **ProxySG2** in its name is associated with this EC2 instance.

|                                     |              |                     |              |          |
|-------------------------------------|--------------|---------------------|--------------|----------|
| <input checked="" type="checkbox"/> | ProxyServer2 | i-0d7e7b543afed7749 | En ejecución | t2.small |
| <input type="checkbox"/>            | ProxyServer1 | i-005ecef81045b42f7 | En ejecución | t2.small |

## i-0d7e7b543afed7749 (ProxyServer2)

#### Rol de IAM

SsmRole

#### Hora de lanzamiento

Fri May 30 2025 13:21:14 GMT-0500 (hora estándar oriental)

#### ID del propietario

137328169203

#### Grupos de seguridad

sg-02e776086f76718ec  
(c162226a4160672110375184t1w137328169203-ProxySG2-9l925tfE7adR)

### ▼ Reglas de entrada

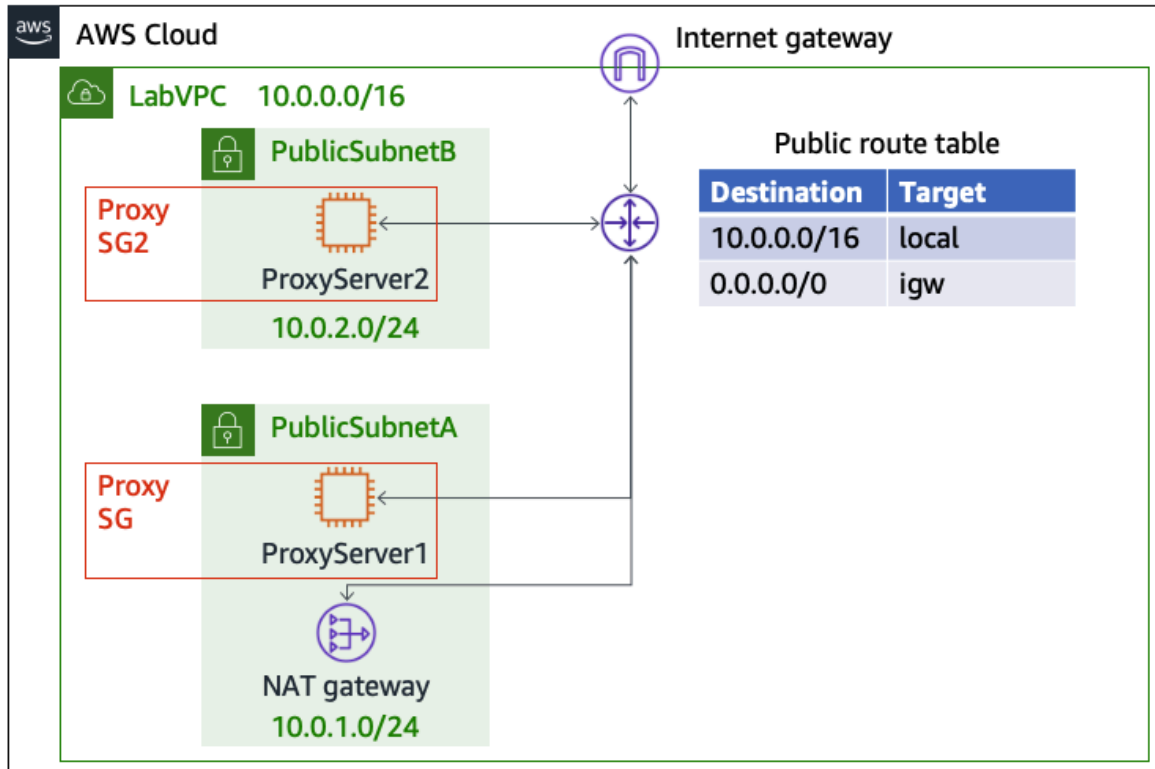
Add a new inbound rule to the ProxySG2 security group.

- Choose the link for the **ProxySG2** security group.
- Choose the **Inbound rules** tab, and then choose **Edit inbound rules**.
- Choose **Add rule**, and configure as follows:
  - Type:** Choose **HTTP**.
  - Source:** Choose **Anywhere-IPv4**.

- Choose **Save rules**.

**Note:** Each security group can be configured differently, even though you just configured the inbound rule settings for ProxySG2 to match the ones found in ProxySG. This means that you are able to independently configure how each instance or group of instances can be accessed.

The following diagram shows the updated subnet routing. ProxyServer1 is associated with ProxySG, while ProxyServer2 is associated with ProxySG2. However, both security groups have the same inbound rules now.



**Note:** When you associate a security group with an instance, you are actually assigning it to an elastic network interface on the instance. Many EC2 instances have one network interface, but an instance can have multiple network interfaces. You can attach a different security group to each network interface on an instance for different access behavior depending on the network connection.

In the next task, you will test connectivity to a webpage that is running on the AppServer. Later in the lab, you will modify security group rules and change which instances they are attached to.

### Task 3: Testing HTTP connectivity from public EC2 instances

In this task, you connect to the different instances that have been provisioned as part of your lab environment. Currently, all three security groups are configured to allow connections over TCP port 80 from all traffic sources.

The proxy servers are located in *public* subnets, which are accessible from the internet. The application server is located in a *private* subnet and is only accessible from servers located inside the VPC, either in a public or private subnet. To connect to the AppServer instance's web server, you will first connect to one of the proxy servers, which are both configured to forward TCP traffic on port 80 to the AppServer.

11. Access the ProxyServer1 public IP address to test loading a webpage that is hosted on the AppServer.

- In the Amazon EC2 console, in the navigation pane, choose **Instances**, and then select the **ProxyServer1** instance.
- On the **Details** tab below, copy the **Public IPv4 address**, and paste it in a new browser tab.



**Hello from the web server running on the AppServer instance!**

The website that loads is actually hosted on the *application server*. You are able to access it because ProxyServer1 is configured to forward all TCP port 80 traffic to the AppServer, and both the ProxySG security group and the AppServerSG security group are configured to allow the inbound traffic.

- Use the same method from the previous step to test loading a webpage by accessing the ProxyServer2 public IPv4 address.



**Hello from the web server running on the AppServer instance!**

The website that loads is the same as the one you opened when you accessed the public IP address for ProxyServer1. This confirms that both ProxyServer1 and ProxyServer2 have access to the web server that is running on the AppServer instance.

13. Close both tabs where the application server website is open.

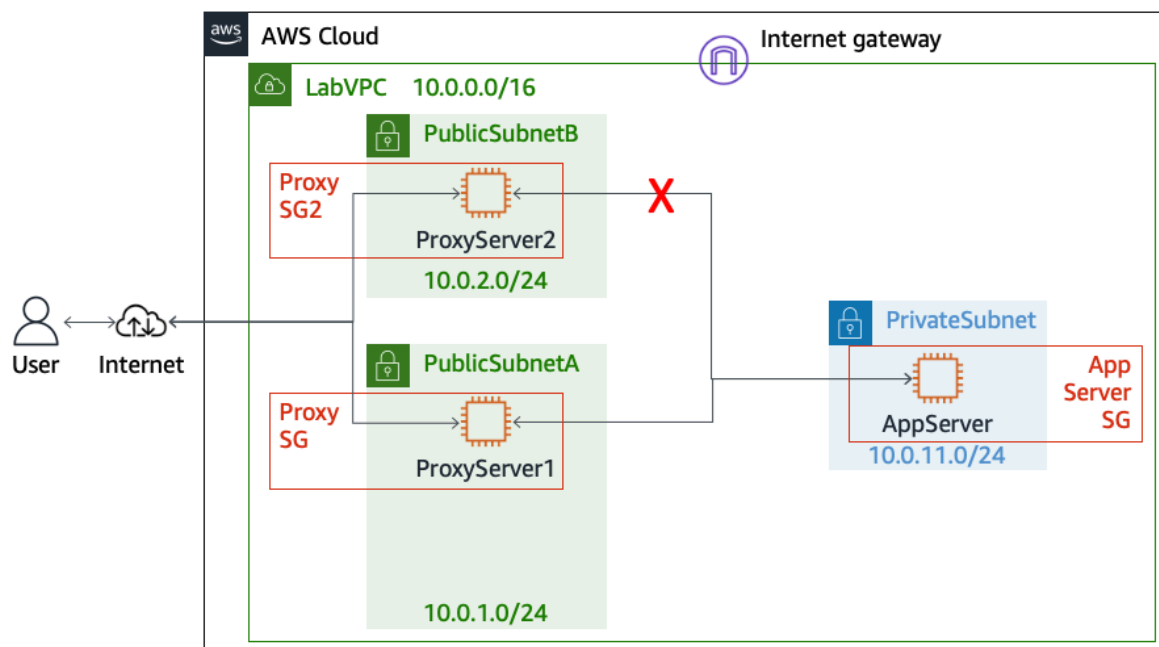
**Note:** Keep the Amazon VPC console and Amazon EC2 console tabs open to use in later steps.

In the next task, you will update the AppServer security group to restrict access to the AppServer.

## Task 4: Restricting HTTP access by using an IP address

In this task, you will update the AppServerSG security group to only allow access from a single internal IP address, as shown in the following diagram. After the update, you will test

access again.



Copy the ProxyServer1 *private* IPv4 address, which you will need to complete the next step.

- You could find this value in the Amazon EC2 console. However, for convenience, you can access this and other environment values that you will need in this lab by choosing the **AWS Details** link above these instructions. )10.0.1.33
- Copy the **ProxyServer1PrivateIP** address to your clipboard.

**Note:** The IP address starts with *10.0.1*.

Edit the AppServerSG security group inbound rules.

- In the Amazon EC2 console, in the navigation pane, choose **Security Groups**.

**Note:** You can access security groups from both the Amazon VPC and Amazon EC2 consoles.

- Select the **AppServerSG** security group.
- Choose the **Inbound rules** tab, and then choose **Edit inbound rules**.
- Modify the **HTTP** rule as follows:
  - In the **Source** field, delete 0.0.0.0/0 by choosing the **X** icon.
  - Next, paste the ProxyServer1PrivateIP value from the lab instructions into the **Source** field.
  - After pasting, add `/32` to the end of the IP address.
- Choose **Save rules**.

You have now updated the security group attached to the application server so that it will *only* allow TCP port 80 traffic from **ProxyServer1**.



**Grupos de seguridad (1/5)** Información

Acciones Exportar los grupos de seguridad a CSV Crear grupo de seguridad

Buscar grupos de seguridad por atributo o etiqueta

|                                     | Name        | ID de grupo de seguridad             | Nombre del grupo de seguridad   |
|-------------------------------------|-------------|--------------------------------------|---------------------------------|
| <input checked="" type="checkbox"/> | AppServerSG | <a href="#">sg-0666eaa18124b4a29</a> | c162226a4160672l10375184t1w1373 |
| <input type="checkbox"/>            | -           | <a href="#">sg-01f7b79453d44576f</a> | default                         |
| <input type="checkbox"/>            | ProxySG     | <a href="#">sg-0b57d1f22ccf750e4</a> | c162226a4160672l10375184t1w1373 |
| <input type="checkbox"/>            | ProxySG2    | <a href="#">sg-02e776086f76718ec</a> | c162226a4160672l10375184t1w1373 |
| <input type="checkbox"/>            | -           | <a href="#">sg-0ede9ba9191265ff4</a> | default                         |

sg-0666eaa18124b4a29 - c162226a4160672l10375184t1w137328169203-AppServerSG-hrfNI91CUABU

< Detalles Reglas de entrada Reglas de salida Compartiendo : novedad >

**Reglas de entrada (1)** Administrar etiquetas Editar reglas de entrada

Buscar

|  | Tipo | Protocolo | Intervalo de puertos | Origen       |
|--|------|-----------|----------------------|--------------|
|  | HTTP | TCP       | 80                   | 10.0.1.33/32 |

16. Test website access again by loading the ProxyServer1 public IP address in a browser tab.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer1PublicIP** value, and then paste it in a new browser tab.

**Note:** If you kept the browser tab open from testing before, perform a hard refresh of the browser tab to ensure that the website is not loading from local cache.

The website that loads is again the application server.

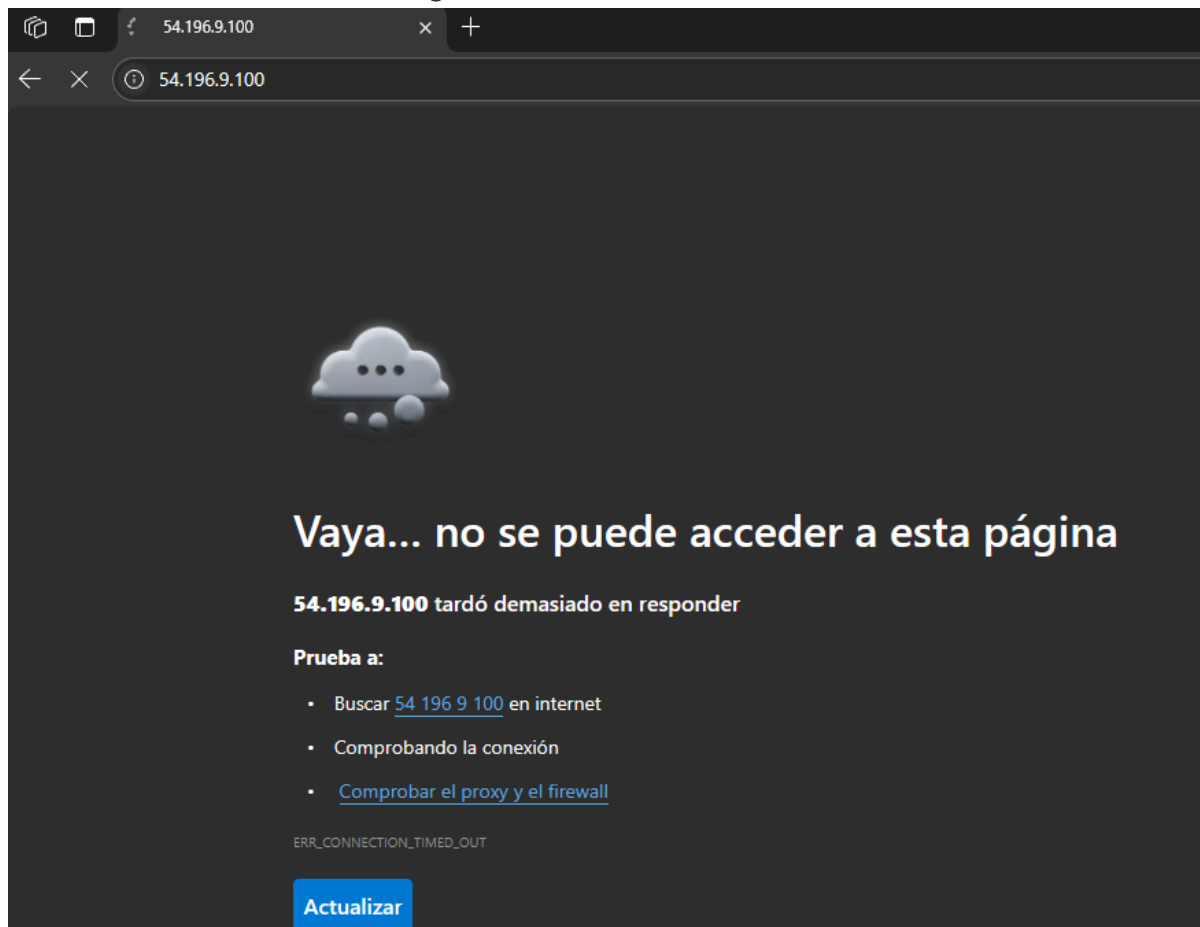
You are still able to access the application server webpage from ProxyServer1 because the server is still forwarding the traffic, and the AppServerSG security group recognizes the source of the request as ProxyServer1.

17. Test website access again by using the ProxyServer2 public IP address.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer2PublicIP(54.196.9.100)** value, and then paste it in a new browser tab.

The website fails to load. If you wait long enough (not a requirement), after about 2 minutes, your browser will display a *connection timeout* error. This is expected because the AppServer's security group only allows traffic from the IP address that is assigned to ProxyServer1 and rejects traffic from any other source.

**Note:** If the website loads, perform a hard refresh of the browser tab to ensure that the website is not loading from local cache.



18. Close both tabs where the application server website is open.

**Note:** Keep the Amazon VPC console and Amazon EC2 console tabs open to use in later steps.

In this task, you modified a security group to only allow incoming access from a single source IP address. In the next task, you will modify the AppServerSG security group to allow traffic from any EC2 instance in the LabVPC that has a specific security group attached.

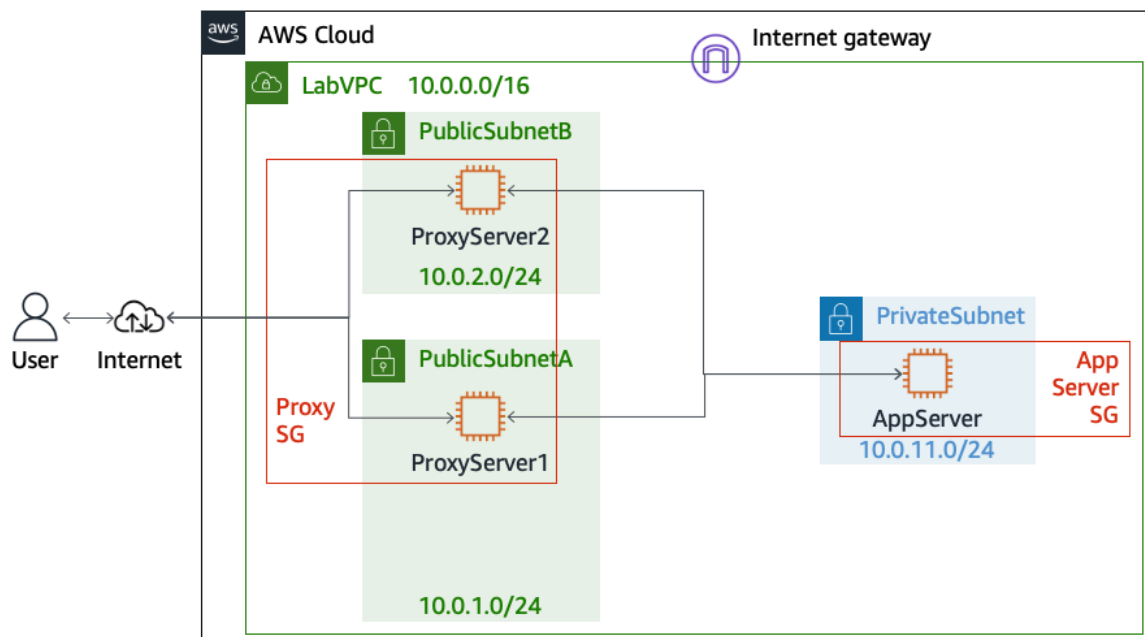
## Task 5: Scaling restricted HTTP access by referencing a security group

You have decided to turn ProxyServer2 into another proxy host to create redundancy. What is the simplest way to give this server and any additional future servers the same access permissions that are currently assigned to ProxyServer1?

You could use an IP address as the source in a security group as demonstrated in the previous task. However, you would need to hardcode the IP address of every instance that will act as a proxy server in the inbound rules of the security group. The simplest, and most secure way, to deploy the same access permissions for multiple hosts is to reference a security group name as the inbound source rather than individual IP addresses.

Up to this point, the lab has focused on the security group associated with the *application server*. Now you will look at the security group assigned to the ProxyServer2 host. You can reference the ProxySG *security group* as the incoming source for HTTP connections to the application server. This means that any instance that is assigned the ProxySG security group will be granted permissions to send HTTP traffic over port 80 to the application server.

The following diagram reflects how the security groups and subnet routing will be set up after you complete this task.

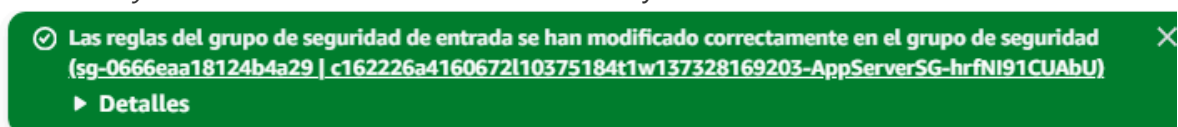


Modify the AppServerSG inbound rules once more so that the allowed source for HTTP traffic is the ProxySG security group instead of the ProxyServer1 internal IP address.

- In the Amazon EC2 console, choose **Security Groups**.
- Select the **AppServerSG** group.
- Choose the **Inbound rules** tab, and then choose **Edit inbound rules**.

- Delete the existing inbound rule that you modified earlier.
- Choose **Add rule**, and configure as follows:
  - **Type:** Choose **HTTP**.
  - **Source:** Choose **Custom**.
  - In the **Source** field, enter `sg`
  - When the dropdown list of existing security groups appears, choose the **ProxySG** security group.
- Choose **Save rules**.

You have now updated the security group attached to the application server so that it will *only* allow traffic from instances with ProxySG attached.



**Grupos de seguridad (1/5)** [Información](#)

[Acciones](#)
[Exportar los grupos de seguridad a CSV](#)
[Crear grupo de seguridad](#)

Buscar grupos de seguridad por atributo o etiqueta

|                                     | Name        | ID de grupo de seguridad             | Nombre del grupo de seguridad                                    |
|-------------------------------------|-------------|--------------------------------------|------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | AppServerSG | <a href="#">sg-0666eaa18124b4a29</a> | c162226a4160672l10375184t1w137328169203-AppServerSG-hrfNI91CUAbU |
| <input type="checkbox"/>            | -           | <a href="#">sg-01f7b79453d44576f</a> | default                                                          |
| <input type="checkbox"/>            | ProxySG     | <a href="#">sg-0b57d1f22ccf750e4</a> | c162226a4160672l10375184t1w137328169203-ProxySG-hrfNI91CUAbU     |
| <input type="checkbox"/>            | ProxySG2    | <a href="#">sg-02e776086f76718ec</a> | c162226a4160672l10375184t1w137328169203-ProxySG2-hrfNI91CUAbU    |
| <input type="checkbox"/>            | -           | <a href="#">sg-0ede9ba9191265ff4</a> | default                                                          |

**sg-0666eaa18124b4a29 - c162226a4160672l10375184t1w137328169203-AppServerSG-hrfNI91CUAbU**

**Reglas de entrada (1)** [Administrar etiquetas](#) [Editar reglas de entrada](#)

Buscar

| Tipo | Protocolo | Intervalo de puertos | Origen                               |
|------|-----------|----------------------|--------------------------------------|
| HTTP | TCP       | 80                   | <a href="#">sg-0b57d1f22ccf750e4</a> |

20. Adjust which security group is attached to the ProxyServer2 instance.

- Still in the Amazon EC2 console, in the navigation pane, choose **Instances**.
- Select the **ProxyServer2** instance.
- Choose the **Actions** menu in the upper-right corner, and then choose **Security** > **Change security groups**.
- Remove the **ProxySG2** security group.
- In the search box, search for and choose the **ProxySG** security group.
- Choose **Add security group**, and then choose **Save**.

Now that you have updated the configuration, test to see which hosts are able to access the application server instance.

### Cambiar grupos de seguridad Información

Amazon EC2 evalúa todas las reglas de los grupos de seguridad seleccionados para controlar el tráfico entrante y saliente de la instancia. Puede utilizar esta ventana para agregar y eliminar grupos de seguridad.

#### Detalles de la instancia

ID de la instancia  
i-0d7e7b543afed7749

ID de la interfaz de red  
eni-06ea183e1586b8af3

#### Grupos de seguridad asociados

Agregue uno o varios grupos de seguridad a la interfaz de red. También puede eliminar grupos de seguridad.

Agregar grupo de seguridad

#### Los grupos de seguridad asociados a la interfaz de red (eni-06ea183e1586b8af3)

| ID de grupo de seguridad | Nombre del grupo de seguridad                                | Descripción                  | ID del propietario |
|--------------------------|--------------------------------------------------------------|------------------------------|--------------------|
| sg-0b57d1f22ccf750e4     | c162226a4160672l10375184t1w137328169203-ProxySG-OJLZeIJADixK | Proxy Server1 Security Group | 137328169203       |

21. Test accessing the website from ProxyServer1.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer1PublicIP** value, and then paste it in a new browser tab.

The website loads.



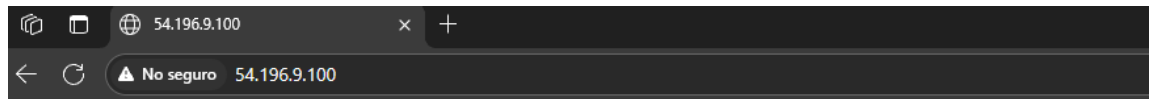
**Hello from the web server running on the AppServer instance!**

22. Test accessing the website from ProxyServer2, which is now in the same security group as ProxyServer1.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer2PublicIP** value, and then paste it in a new browser tab.

The website loads.

**Note:** If the website does not load, perform a hard refresh of the browser tab.



**Hello from the web server running on the AppServer instance!**

23. Close both tabs where the application server website is open.

**Note:** Keep the Amazon VPC console and Amazon EC2 console tabs open to use in later steps.

In this task, you updated the AppServerSG security group to allow traffic from any instances associated with the ProxySG security group. Then, you updated the security group attached to the ProxyServer2 instance to be ProxySG, which is also attached to the ProxyServer1 instance. Therefore, both proxy servers can now connect to the application server, and you don't need to maintain a list of approved source IP addresses in the security group settings.

## Task 6: Restricting HTTP access by using a network ACL

### Network Access Control List

Security groups are not the only method that you can use to secure access to network ports. In this task, you will explore how to use network ACLs to control access to EC2 instances in a VPC.

Add a new inbound rule on the LabVPC network ACL to deny all inbound traffic on port 80.

- Return to the browser tab where the Amazon VPC console is open.
- In the navigation pane, choose **Network ACLs**.
- Select the network ACL that is associated with LabVPC.
- Choose the **Inbound rules** tab, and then choose **Edit inbound rules**
- Choose **Add new rule**, and configure as follows:
  - **Rule number:** Enter 99
  - **Type:** Choose **HTTP**.
  - **Allow/Deny:** Choose **Deny**.
  - Choose **Save changes**.

**Editar reglas de entrada** Información

Las reglas de entrada controlan el tráfico entrante que tiene permiso para llegar a la VPC.

**Regla de entrada 1**

Número de regla Información

99

Tipo Información

HTTP (80)

Protocolo Información

TCP (6)

Rango de puertos Información

80

Origen Información

0.0.0.0/0

Permitir/denegar Información

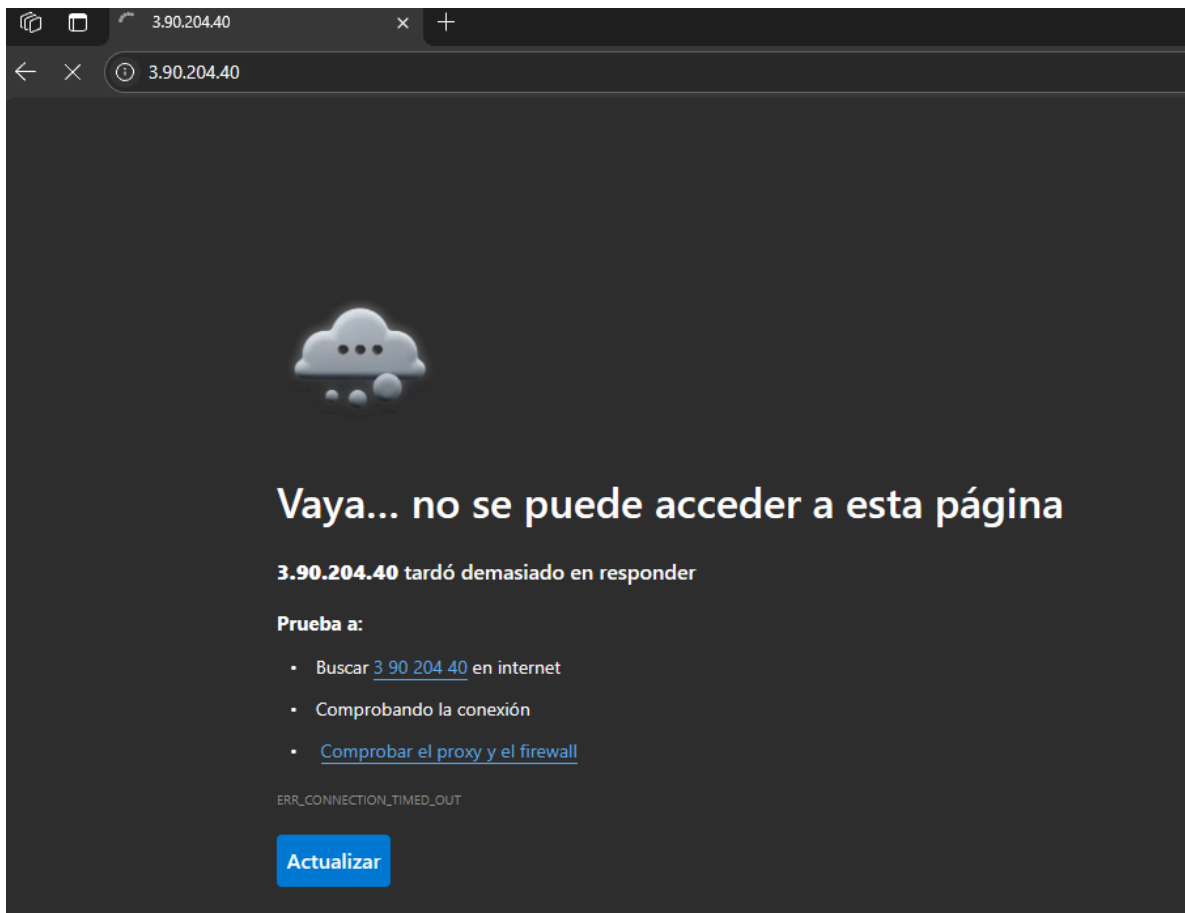
Denegar

Quitar

Test accessing the website from ProxyServer1.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer1PublicIP** value, and then paste it in a new browser tab.

After a few moments, the connection times out.



- **Note:** If the website loads, perform a hard refresh of the browser tab.

**Analysis:** The network ACL rule that you created blocks all connection attempts on port 80 to any EC2 instance running in the VPC. The ProxySG security group allows inbound port 80 traffic, but the connection fails because both the network ACL and security group would need to allow it.

- Close the tab where you tried to load the application server website.

26. Add another rule to the network ACL.

- In the Amazon VPC console, with the LabVPC network ACL still selected, choose **Edit inbound rules**.
- Choose **Add new rule** and configure as described:
  - **Rule number:** Enter 98
  - **Type:** Choose **HTTP**.
  - **Allow/Deny:** Choose **Allow**.
  - Choose **Save changes**.

**Editar reglas de entrada** [Información](#)

Las reglas de entrada controlan el tráfico entrante que tiene permiso para llegar a la VPC.

| Regla de entrada 1                                 | Tipo <a href="#">Información</a>                | Protocolo <a href="#">Información</a>                      |
|----------------------------------------------------|-------------------------------------------------|------------------------------------------------------------|
| Número de regla <a href="#">Información</a><br>98  | HTTP (80) ▼                                     | TCP (6) ▼                                                  |
| Rango de puertos <a href="#">Información</a><br>80 | Origen <a href="#">Información</a><br>0.0.0.0/0 | Permitir/denegar <a href="#">Información</a><br>Permitir ▼ |

[Quitar](#)

Test accessing the website from ProxyServer1 again.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer1PublicIP** value, and then paste it in a new browser tab.

The website loads.



- **Analysis:** All rules defined in a network ACL have a *rule number*. Rules are evaluated in order, starting with the lowest numbered rule. When a rule matches traffic, it's applied regardless of any higher numbered rule that might contradict it.
- Close the tab where you loaded the application server website.

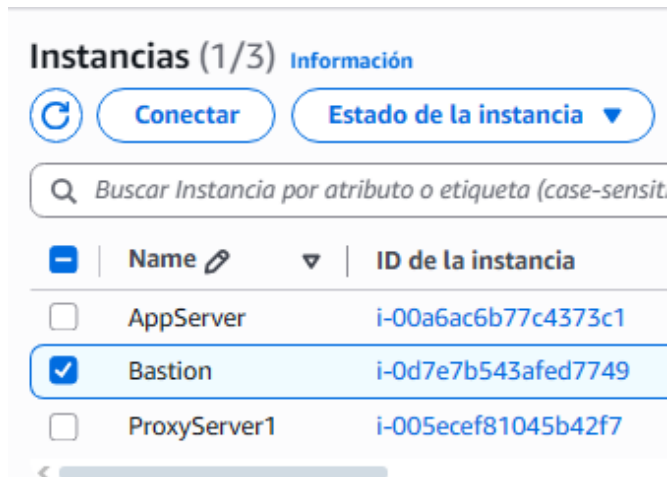


## Task 7: Connecting to the AppServer by using a bastion host and SSH

In this next task, you will begin to explore how you can connect to the AppServer terminal. In this task, you will use an SSH key to connect to a server in PublicSubnetB by using SSH over TCP port 22. Then, by using SSH port forwarding, you will then be able to connect to the AppServer, which is running in the private subnet.

Rename the ProxyServer2 EC2 instance.

- In the Amazon EC2 console, choose **Instances**, and then select the **ProxyServer2** instance.
- In the **Name** column, place your cursor over **ProxyServer2**, and then choose the edit icon that appears.
- Rename the server as **Bastion** and choose **Save**.



**Note:** A *bastion host* (also called a *jump server*) is a server that provides access to a private network from an external network, such as the internet. In a real-world scenario, you would launch a new security-hardened EC2 instance to act as a bastion host. However, for the sake of demonstration in this lab, you will repurpose ProxyServer2 to become a bastion host.

Create a new security group with port 22 open for the bastion host.

- In the navigation pane, choose **Security Groups**.
- Choose **Create security group**, and configure as follows:
  - **Security group name:** Enter **BastionSG**
  - **Description:** Enter **BastionSG**
  - **VPC:** Remove the currently selected VPC by choosing the **X** icon. Then, choose **LabVPC**.
  - In the **Inbound rules** section, choose **Add rule**, and configure as follows:
    - **Type:** Choose **SSH**.

- **Source:** Choose **Anywhere-IPv4**.
- At the bottom of the page, choose **Create security group**.

### Crear grupo de seguridad [Información](#)

Un grupo de seguridad actúa como un firewall virtual para que la instancia controle el tráfico de entrada y salida. Para crear un nuevo grupo de seguridad, complete los campos siguientes.

#### Detalles básicos

**Nombre del grupo de seguridad** [Información](#)

El nombre no se puede editar después de su creación.

**Descripción** [Información](#)

**VPC** [Información](#)

#### Reglas de entrada [Información](#)

Regla de entrada 1 Eliminar

|                                                                                                 |                                                                                     |                                                                                            |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Tipo</b> <a href="#">Información</a><br><input type="text" value="SSH"/>                     | <b>Protocolo</b> <a href="#">Información</a><br><input type="text" value="TCP"/>    | <b>Intervalo de puertos</b> <a href="#">Información</a><br><input type="text" value="22"/> |
| <b>Tipo de origen</b> <a href="#">Información</a><br><input type="text" value="Anywhere-IPv4"/> | <b>Origen</b> <a href="#">Información</a><br><input type="text" value="0.0.0.0/0"/> | <b>Descripción: opcional</b> <a href="#">Información</a><br><input type="text"/>           |

Agregar regla

⚠ Las reglas cuyo origen es 0.0.0.0/0 o ::/0 permiten a todas las direcciones IP acceder a la instancia. Recomendamos configurar reglas de grupo de seguridad para permitir el acceso únicamente desde direcciones IP conocidas. ✕

30. Update the bastion host instance settings to use the BastionSG security group.

- In the navigation pane, choose **Instances**.
- Select the **Bastion** instance.
- Choose **Actions** > **Security** > **Change security groups**.
- Remove the **ProxySG** security group.
- In the search box, search for and choose the **BastionSG** security group.
- Choose **Add security group**, and then choose **Save**.

## Cambiar grupos de seguridad [Información](#)

Amazon EC2 evalúa todas las reglas de los grupos de seguridad seleccionados para controlar el tráfico entrante y saliente de la instancia. Puede utilizar esta ventana para agregar y eliminar grupos de seguridad.

### Detalles de la instancia

ID de la instancia  
 i-0d7e7b543afed7749

ID de la interfaz de red  
 eni-06ea183e1586b8af3

### Grupos de seguridad asociados

Agregue uno o varios grupos de seguridad a la interfaz de red. También puede eliminar grupos de seguridad.



[Agregar grupo de seguridad](#)

#### Los grupos de seguridad asociados a la interfaz de red (eni-06ea183e1586b8af3)

| ID de grupo de seguridad | Nombre del grupo de seguridad | Descripción | ID del propietario |                          |
|--------------------------|-------------------------------|-------------|--------------------|--------------------------|
| sg-03a163230640f35dd     | BastionSG                     | BastionSG   | 137328169203       | <a href="#">Eliminar</a> |

BASTION PRIVATE IP 10.0.2.120

31. Edit the inbound rules for the AppServerSG security group to allow traffic from the bastion host.

- Choose the **AWS Details** link above these instructions, and copy the **BastionPrivateIP** value.
- In the Amazon EC2 console, in the navigation pane, choose **Security Groups**.
- Select the **AppServerSG** security group.
- Choose the **Inbound rules** tab, and then choose **Edit inbound rules**.
- Choose **Add rule**, and configure as follows:
  - **Type:** Choose **SSH**.
  - **Source:** Choose **Custom**.
  - Next, paste the BastionPrivateIP value from the lab instructions into the **Source** field.
  - After pasting, add **/32** to the end of the IP address.
- Choose **Save rules**.

You have now updated the security group that is attached to the application server so that it will allow SSH connections on port 22 from the bastion host.

## Editar reglas de entrada [Información](#)

Las reglas de entrada controlan el tráfico entrante que puede llegar a la instancia.

### Reglas de entrada [Información](#)

**Regla de entrada 1** Eliminar

**ID de la regla del grupo de seguridad**  
sgr-01037c7015ada65a8

**Tipo** [Información](#)  
HTTP ▼

**Protocolo** [Información](#)  
TCP

**Intervalo de puertos** [Información](#)  
80

**Tipo de origen** [Información](#)  
Personalizada ▼

**Origen** [Información](#)  
Q sg-0b57d1f22ccf750e4 X

**Descripción: opcional** [Información](#)

**Regla de entrada 2** Eliminar

**ID de la regla del grupo de seguridad**  
-

**Tipo** [Información](#)  
SSH ▼

**Protocolo** [Información](#)  
TCP

**Intervalo de puertos** [Información](#)  
22

**Tipo de origen** [Información](#)  
Personalizada ▼

**Origen** [Información](#)  
Q 10.0.2.120/32 X  
10.0.2.120/32 X

**Descripción: opcional** [Información](#)

Agregar regla

Connect to the bastion host by using SSH.

- In the terminal panel next to these instructions, run the following two commands.

```
exec ssh-agent bash
ssh-add ~/.ssh/labsuser.pem
```

**Note:** The first command activates the ssh-agent tool on this Ubuntu machine, which exists outside the lab AWS account. The second command adds the SSH key to the agent. Both the bastion host and AppServer will accept the labsuser.pem SSH key for SSH connection requests.

BASTION PUBLIC IP 54.196.9.100

Next, to connect to the bastion instance, run the following command. Replace *<BastionPublicIP>* with the **BastionPublicIP** value, which you can find by choosing the **AWS Details** link above these instructions.

```
ssh -i ~/.ssh/labsuser.pem -A ec2-user@<BastionPublicIP>
```

- When prompted if you are sure you want to connect, enter `yes`

**Tip:** Notice that the command prompt now reads `[ec2-user@bastion]` which indicates that you are now connected to the bastion instance.

```
PERMISSION DENIED (publickey,gssapi-keyex,gssapi-with-mic).
eee_W_4462054@runweb175309:~$ ssh -i .ssh/labsuser.pem -A ec2-user@54.196.9.100
_#_
~\##### Amazon Linux 2
~\#####\
~\#####| AL2 End of Life is 2026-06-30.
~\#####|
~\#####| \#/_
~\#####| V~' '->
~\#####|
~\#####| A newer version of Amazon Linux is available!
~\#####|
~\#####| Amazon Linux 2023, GA and supported until 2028-03-15.
~\#####| https://aws.amazon.com/linux/amazon-linux-2023/
~\#####|
_/m/'

[ec2-user@bastion ~]$
```

Connect to the AppServer through SSH by forwarding the authenticated connection.

- Run the following command. Replace `<AppServerPrivateIP>` with the **AppServerPrivateIP** value, which you can find by choosing the **AWS Details** link above these instructions.
- BASTION PRIVATE IP 10.0.2.120

```
ssh ec2-user@<AppServerPrivateIP>
```

- When prompted if you are sure you want to connect, enter `yes`

You are now connected to the AppServer.

**Tip:** Notice that the command prompt now reads `[ec2-user@appserver]` which indicates that you are now connected to the AppServer instance.

```
[ec2-user@bastion ~]$ ssh ec2-user@10.0.2.120
The authenticity of host '10.0.2.120 (10.0.2.120)' can't be established.
ECDSA key fingerprint is SHA256:b01G9Mh255qhPSXn7gt0jHmswmdf4Zwt/OqHLcoafWA.
ECDSA key fingerprint is MD5:98:3c:73:31:61:03:d7:c6:ac:f4:c0:d0:19:19:05:13.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.0.2.120' (ECDSA) to the list of known hosts.
Last login: Fri May 30 20:23:14 2025 from ec2-54-202-133-19.us-west-2.compute.amazonaws.com
_#_
~\##### Amazon Linux 2
~\#####\
~\#####| AL2 End of Life is 2026-06-30.
~\#####|
~\#####| \#/_
~\#####| V~' '->
~\#####|
~\#####| A newer version of Amazon Linux is available!
~\#####|
~\#####| Amazon Linux 2023, GA and supported until 2028-03-15.
~\#####| https://aws.amazon.com/linux/amazon-linux-2023/
~\#####|
_/m/'

[ec2-user@bastion ~]$
```

To create a file to prove that you connected to the AppServer instance, run the following command. (The presence of this file will be verified when you submit the lab.)

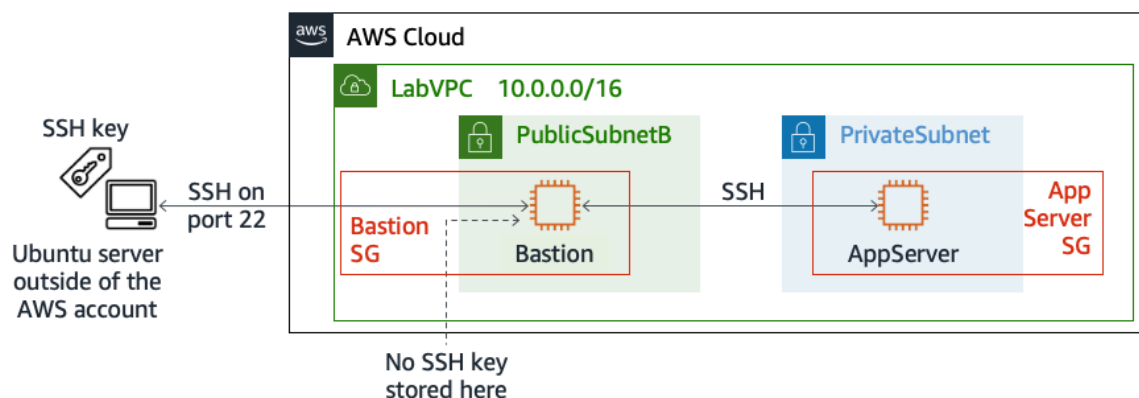
```
touch newfile.txt
```

**Detailed analysis:** You were able to connect to a server in a private subnet because, when you issued the `ssh` command from the Ubuntu server outside the account, you included `-A` in the command. This enables the authenticated agent connection to be forwarded.

The `labsuser.pem` SSH key is not stored on the bastion host, but the SSH session that you successfully opened from the outside server to the bastion host included it. The AppServer's SSH settings are configured to accept the same `labsuser.pem` SSH key.

Furthermore, the connection to the AppServer was made *by way of the bastion host*, which you configured in the inbound rules of the AppServerSG security group as an acceptable source for SSH connections. Therefore, the connection was successful. Importantly, by using this approach, you avoided storing the SSH key on the bastion host, which is not a good security practice.

The following diagram shows the configuration that you just configured and used.



34. In the terminal, to disconnect from the AppServer, run the `exit` command.

**Tip:** Notice that the command prompt now reads `[ec2-user@bastion]` which indicates that you dropped the connection to the AppServer and are again back on the bastion instance.

35. To disconnect from the bastion server, run the `exit` command.

Notice that the prompt now shows you are connected to the Ubuntu terminal (as indicated by `@runweb` in the prompt). The Ubuntu server exists outside the AWS account.

Could you connect to an EC2 instance in a VPC, even an instance in a private subnet, without opening the SSH port? In the next task, you will explore how to use Systems

Manager Session Manager to access instances in private subnets without having to use a bastion host.

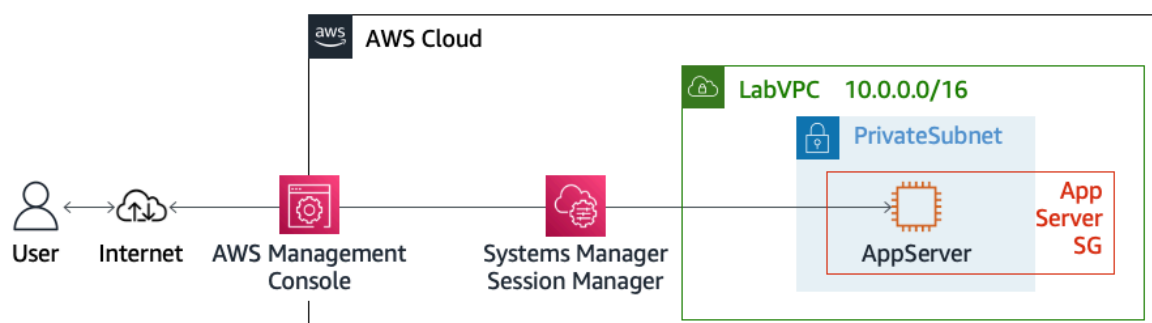
## Task 8: Connecting directly to a host in a private subnet by using Session Manager

What should you do if you need to modify configuration settings for your application server? You would need to connect to one of the servers in a *public* subnet (often called a bastion host or jump box) and use that to connect to the server in the *private* subnet.

Because your instances are located in the AWS Cloud, you can use the Systems Manager Session Manager to connect directly to an instance *without* opening port 22 for SSH or connecting through another server in a public subnet first. For this to work, the Session Manager agent must be installed on the target host. Amazon Linux EC2 instances already have the Session Manager agent installed. You can connect with Session Manager by using a URL through a browser, as you have been doing in this lab, or through the AWS Command Line Interface (AWS CLI).

Using a bastion host is a best practice in a VPC network, but using Session Manager can be an alternative to setting up and maintaining a bastion host. By using Session Manager, AWS is doing the heavy lifting of securing the connection and managing a secure, redundant bastion host configuration. This also gives you the ability to use AWS Identity and Access Management (IAM) to grant access to users, and creates an auditable logging trail when integrated with AWS CloudTrail and Amazon CloudWatch.

In this task, you will directly connect to the application instance, which is not internet accessible. You will use Session Manager, as shown in the following diagram.



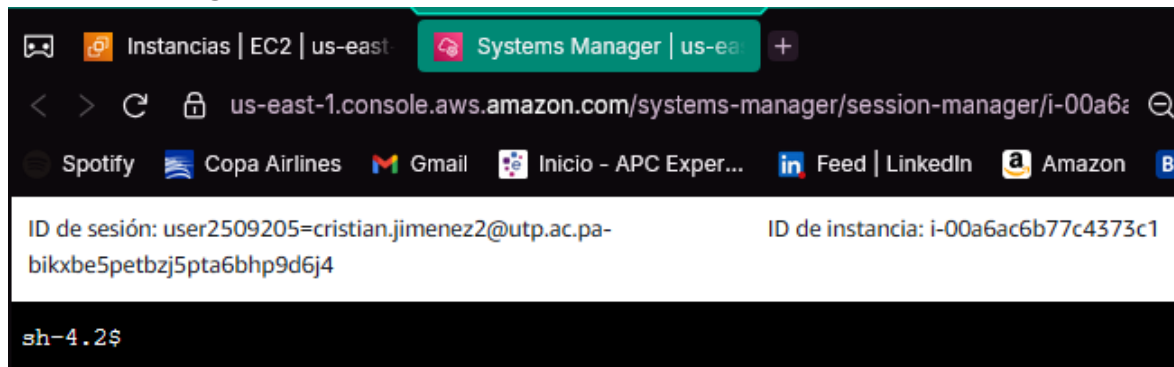
Connect to the AppServer by using Session Manager.

- In the browser tab where the Amazon EC2 console is open, choose **Instances**.

- Select **AppServer**, and choose **Connect** in the upper-right corner.
- Choose the **Session Manager** tab, and then choose **Connect**.

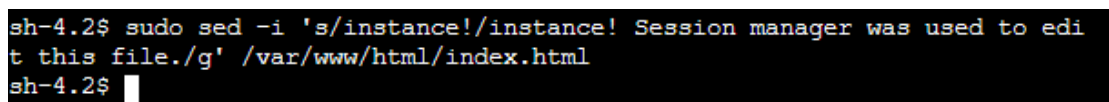
A new browser tab opens, and you are connected to the AppServer terminal with a `sh-4.2` prompt.

You are now connected into an SSH session on the application server without having to use a bastion host. Recall also that the network ACL rule blocking traffic on port 22 is still configured.



To modify the website hosted on this server, run the following command:

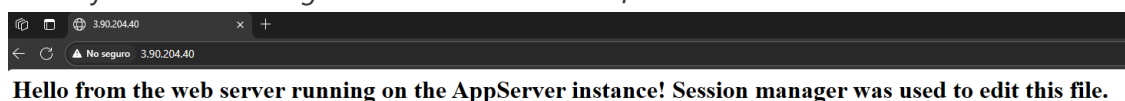
```
sudo sed -i 's/instance!/instance! Session manager was used to edit this file./g' /var/www/html/index.html
```



Test access to the webpage hosted on the AppServer again, by using the ProxyServer1 public IP address.

- Choose the **AWS Details** link above these instructions, copy the **ProxyServer1PublicIP** value, and then paste it in a new browser tab.

The application server website loads. Notice that the application server website now says *Session manager was used to edit this file*.



**Note:** If necessary, perform a hard refresh of the browser tab to ensure that the website is not loading from local cache.

Congratulations! You successfully connected to an instance that is located in a private subnet and modified a file on that instance securely. By using this approach, you can avoid



the many detailed configuration steps that were required to use a bastion host to connect.